CMPS 460 Machine Learning Project – Spring 2026
Due by 11:59pm on Monday, 4th May 2026

The goal of this project is to apply the complete machine learning lifecycle to a real-world dataset, including data exploration, preprocessing, model training, evaluation, and optimization.
Deliverable: One well-organized **Jupyter Notebook** per team that includes:

1. Dataset Selection

- Select a realistic, non-trivial dataset from sources such as Kaggle, public repositories, or private data that you have permission to access.
- Ensure the dataset is suitable for supervised machine learning.

2. Exploratory Data Analysis (EDA)

- Describe dataset structure and features.
- Use summary statistics and meaningful visualizations, then draw meaningful observations.
- Identify distributions, correlations, patterns, or anomalies.

3. Data Cleaning and Preprocessing

- Handle missing values, outliers, and duplicates.
- Perform feature engineering guided by correlation analysis.
- Split data into training and testing sets.
- For a large dataset, you may use stratified sampling when appropriate.

4. Model Design and Training

Train **at least three models**, each from a different category:

- **Traditional ML Model** (e.g., Linear Regression, Decision Tree, Naïve Bayes)
- **Ensemble Model** (e.g., Random Forest, XGBoost)
- **Deep Learning Model** (e.g., Neural Network)

For each model:

- Justify why it is appropriate for the dataset.
- Describe architecture and key parameters used.

5. Model Evaluation and Comparison

- Evaluate models using appropriate metrics (e.g., accuracy, F1-score, MSE).
- Compare performance across models.
- Analyze strengths, weaknesses, and error patterns.

6. Model Optimization

- Apply optimization techniques such as:
  - Hyperparameter tuning
  - Regularization
  - Feature selection or ranking
- Demonstrate and discuss performance improvements.

7. Documentation and Reporting

Your Jupyter Notebook must include:

- Clear section headings and logical flow
- Well-commented code and explanatory Markdown cells
- Justification for all major decisions
- Interpretation of results and final conclusions

Oral Discussion

Each team **must** attend a **10-minute discussion** during office hours to present the project and answer questions to confirm understanding and authorship.

Project Grading Rubrics

Requirement	Poor (0-50%)	Fair (51-65%)	Good (66-85%)	Excellent (86-100%)
Exploratory Data Analysis (EDA) (**10%**)	- Minimal EDA; few or no meaningful statistics or visualizations. - Little understanding of the dataset's structure, distributions, and relationships. - Minimal or absent interpretation of findings; lacks depth and thoroughness.	- Basic EDA with some summary statistics and visualizations. - Partial understanding of the dataset's structure, distributions, and relationships. - Limited or basic interpretations of findings.	- Good EDA using appropriate summary statistics and visualizations. - Clear understanding of the dataset's structure, distributions, and relationships. - Reasonable interpretations of findings including patterns or anomalies based on the analysis.	- Comprehensive EDA with well-chosen summary statistics and clear visualizations. - Deep insight into the dataset's structure, distributions, and relationships. - Strong interpretation of findings including patterns or anomalies based on the analysis.
Data Cleaning and Preprocessing (**10%**)	- Little to no addressing of missing values, outliers, and duplicates in the dataset. - Limited or simplistic feature engineering.	- Basic handling of missing values, outliers, and duplicates. - Weak feature engineering or lack depth and justifications.	- Effectively handling of missing values, outliers, and duplicates using mostly appropriate techniques. - Reasonable feature engineering with justifications.	- Robust and well-justified preprocessing using appropriate techniques. - Thorough feature engineering with clear justifications.
Model Design and Training (**35%**)	- Models poorly implemented or incomplete; missing required model categories. - Little or no justification. - Little or no explanation of architectures and parameters.	- Two or three models trained but lacking depth, correctness, or justification; weak alignment with dataset. - Weak explanation of architectures and parameters.	- Correct implementation of all three model types; clear and reasonable justification based on dataset and task. - Reasonable explanation of architectures and parameters.	- Expertly designed and implemented models; insightful justification. - Well-explained architectures and parameters.
Model Evaluation and Comparison (**20%**)	- Inadequate or incorrect evaluation. - Little or no comparison between models. - Limited or no analysis of strengths and weaknesses of each model.	- Basic evaluation using limited metrics. - Basic comparison and analysis of results. - Basic analysis of strengths and weaknesses of each model.	- Complete model evaluation using appropriate metrics. - Clear comparison and meaningful analysis of results. - Detailed analysis of strengths and weaknesses of each model.	- Comprehensive evaluation with multiple metrics. - Deep analysis of performance, errors, and trade-offs across models. - In-depth analysis of strengths and weaknesses of each model.
Model Optimization (**20%**)	Little or no optimization attempted; no measurable improvement.	Basic optimization applied; moderate improvement with limited explanation.	Appropriate optimization techniques applied (e.g., tuning, regularization); clear performance gains.	Multiple optimization strategies applied effectively; superior performance with strong technical justification.
Notebook Quality & Organization (**5%**)	Disorganized notebook; unclear flow; poorly commented or unreadable code.	Basic structure present; some comments and explanations but lacking clarity or consistency.	Well-organized notebook; clear sections; readable, well-commented code and explanations.	Exceptionally organized and professional; excellent clarity, documentation, and presentation throughout.